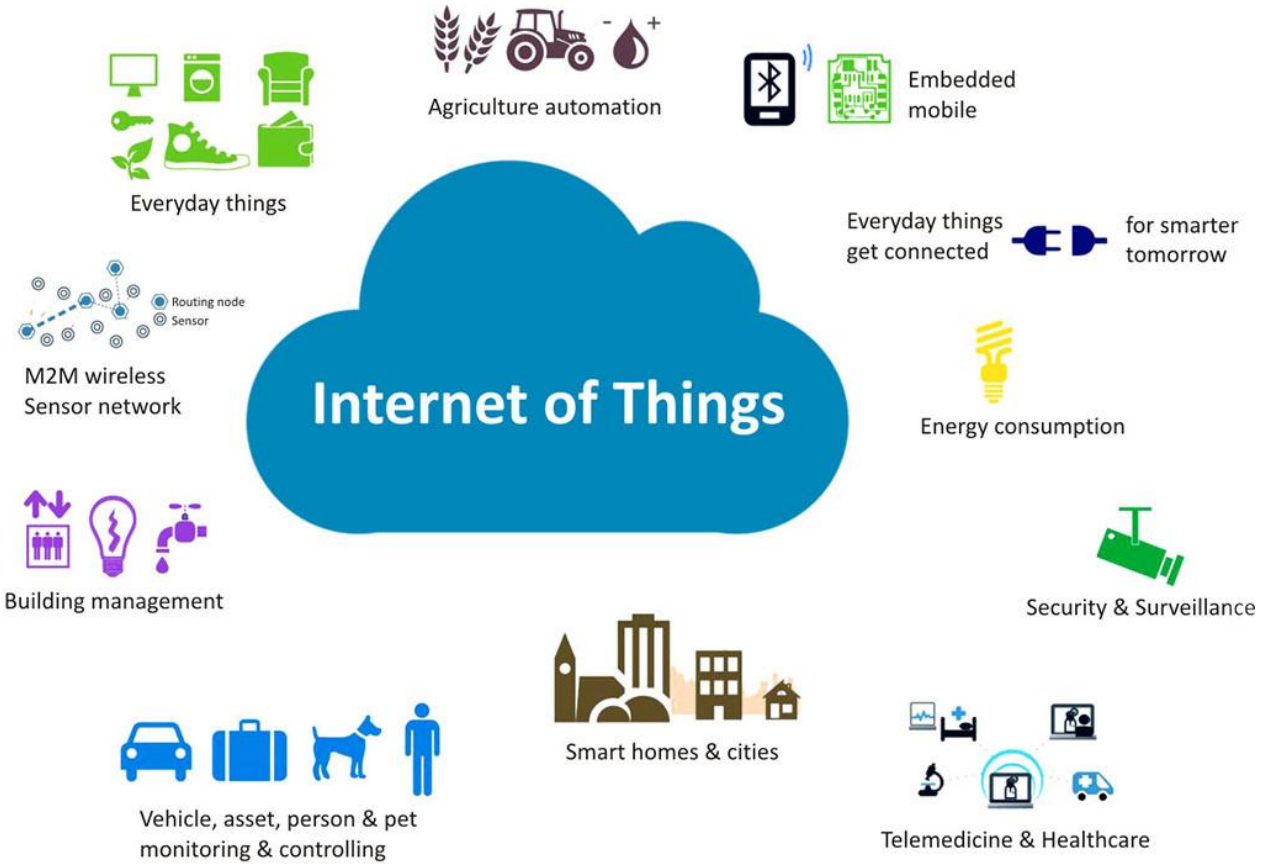
Semantic Sensor Web For Geospatial IoT

MTP
Under the Guidance of
Prof. Surya S. Durbha
CSRE, IIT Bombay

Presented By
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163316001
MTech RA, 3rd Year
CSRE, IIT Bombay

Internet of Things

- ★ Network of Sensors
- ★ Dashboards for visualization
- ★ Analysis, Inferences
- ★ Predictions, Detection of Problems
- ★ Trends and patterns
- ★ Gain insights from the IoT data



Source: https://iot.xtendbusiness.com/smart-homes/ top-internet-things-application-areas/

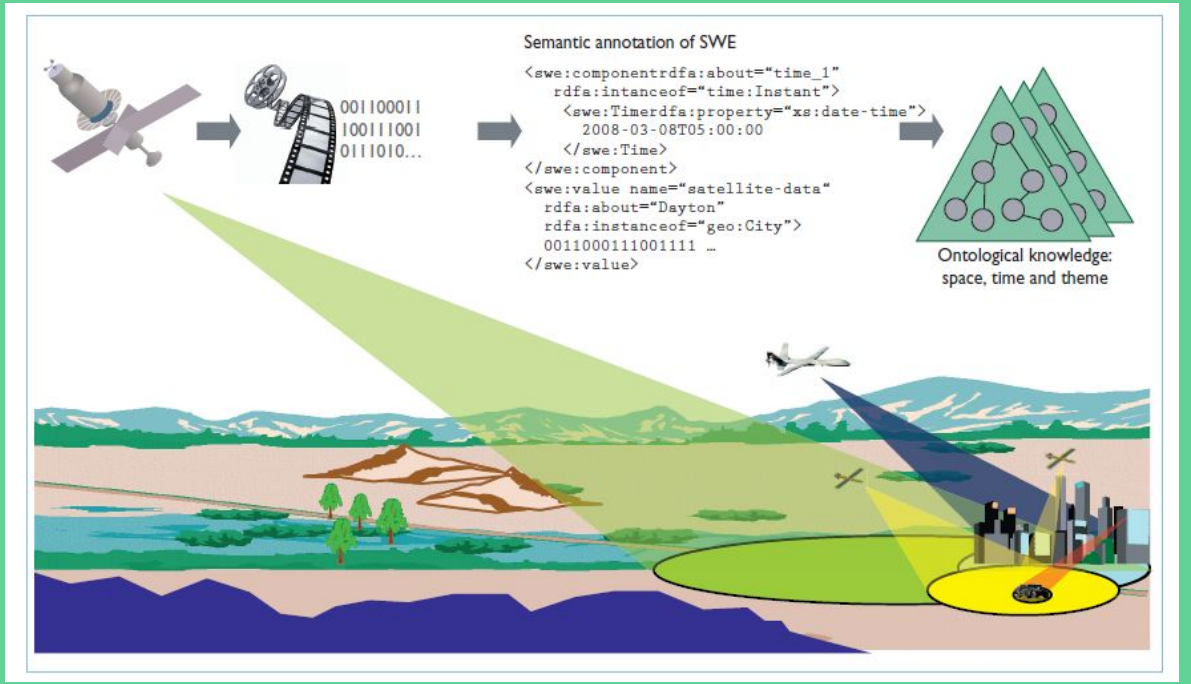
IoT Applications

- ★ Emergency situation - Flood management
- ★ Weather stations, Transport network, Surveillance cameras, Social media like Twitter
- ★ Integration of data from different sources is necessary for real world scenarios
- ★ Interoperability
- ★ Ability to be used across applications, platforms, Operating systems
- ★ OGC Standardization - Data models, Service Models
- ★ Easy discovery and access

Contextual meaning of Data???

Semantic Sensor Web

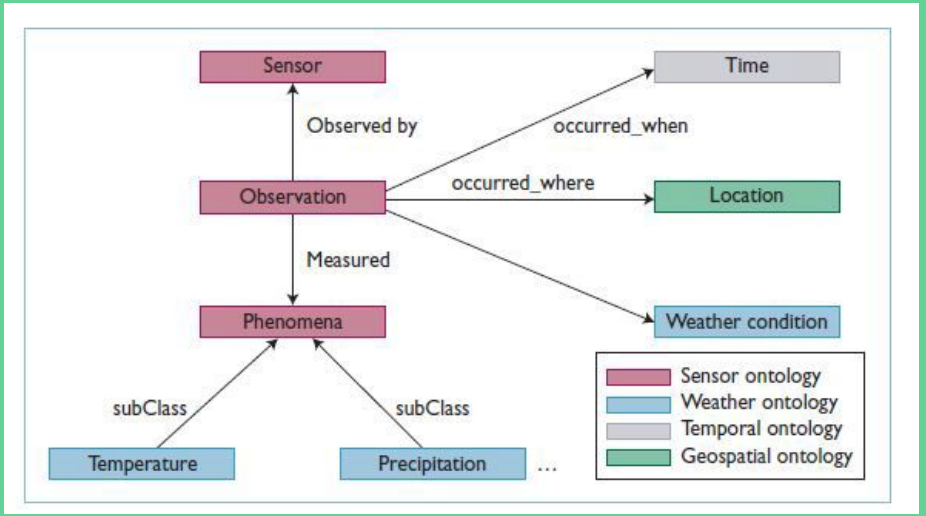
- ★ Meaning or context
- ★ Method to share the meaning of the data
- ★ Define all the implicit logic
- ★ Incorporate Domain knowledge
- ★ Machine understable
- ★ Cross-domain applications



Ref: Semantic Sensor Web, Sheth and Sahoo (2008)

Ontologies

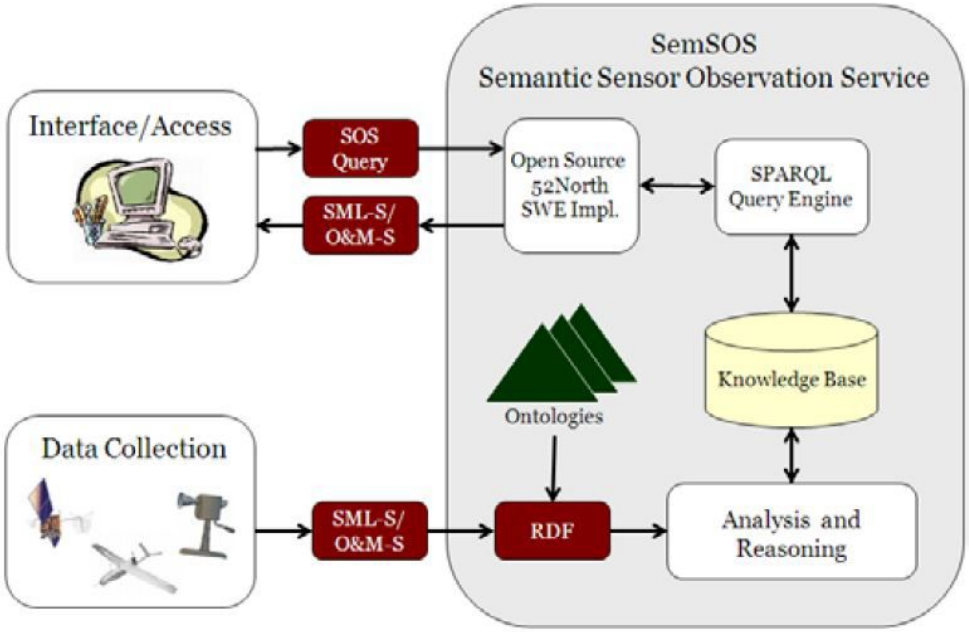
- Formally define domain knowledge
- Classes , Subclasses, Properties, Sub properties
- Method of defining classes and the properties or links between classes
- Supports reasoning – inferences
- Makes implicit knowledge explicit
- Ensure consistency, integration from various data sources



Ref: Semantic Sensor Web, Sheth and Sahoo (2008)

Semantic Sensor Observation Service

- Semantic annotation using O&M in OWL
- Sensor Observation Service
- Complex querying over simple weather data
- Rule based reasoning - infer weather phenomenon
- High Winds, Low Visibility, Snowfall, Blizzards



Ref: Semantic sensor observation service, C. A. Henson et al. (2009)

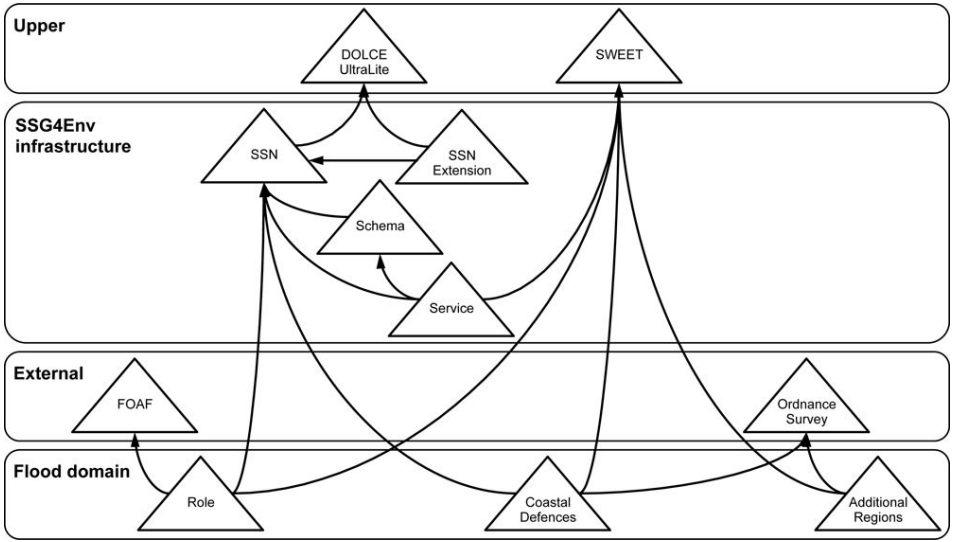
Semantic based applications

- Machine understandable semantics
- Integration of heterogeneous data
- Domains
- Ontologies
- Inferences and Reasoning
- Context based queries
- Assistance agents

State of the Art Semantic Systems

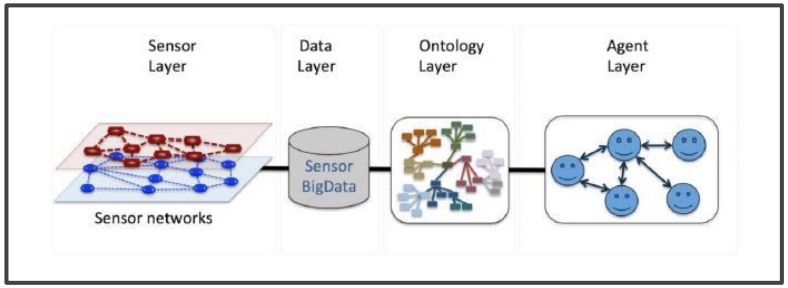
Environmental Decision Support Application using SSW

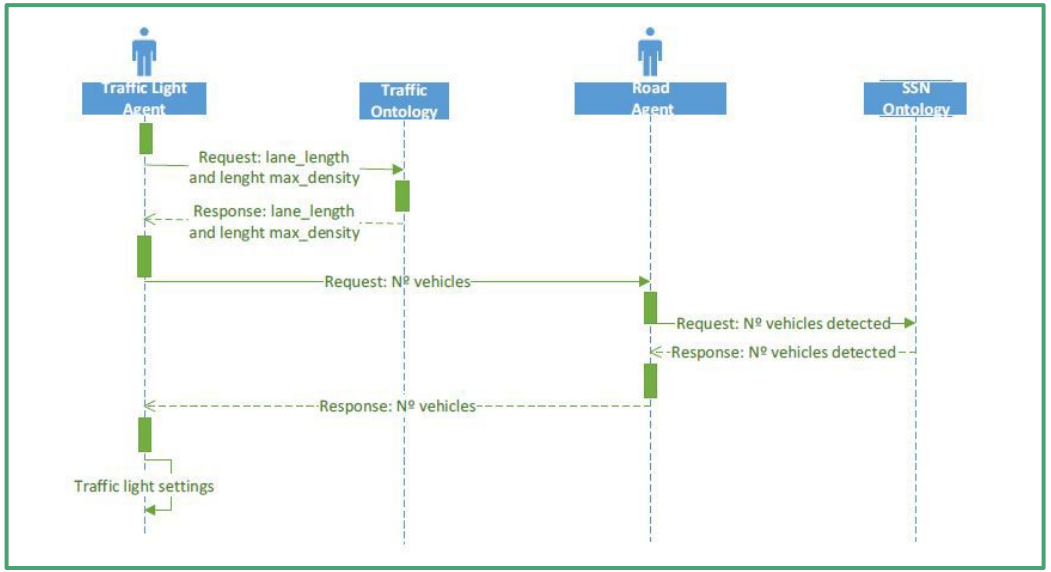
- Flood response planning Web application
- Integrate multiple heterogeneous datasets
- Live and historic data
- Databases, Map Layers
- Discovers dataset using spatio-temporal thematic coverages
- Publish result through HTTP



Ref: Gray et al.,A semantic sensor web for environmental decision support applications (2011)

Intelligent Transport Systems



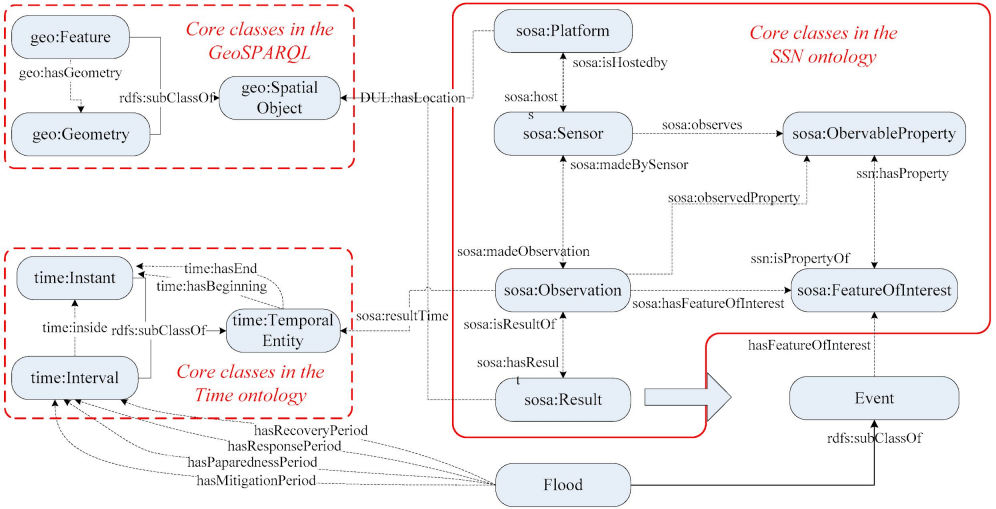


- Sensors - traffic flow, weather and vehicle related
- Ontology - Vehicles, Road infrastructure, Sensors
- W3C SSN-XG
- Agents - Driver Personal Agents, Traffic light agent, Road Agent

Ref: Fernandez et al., Ontology-based architecture for intelligent transportation systems using a traffic sensor network. (2018)

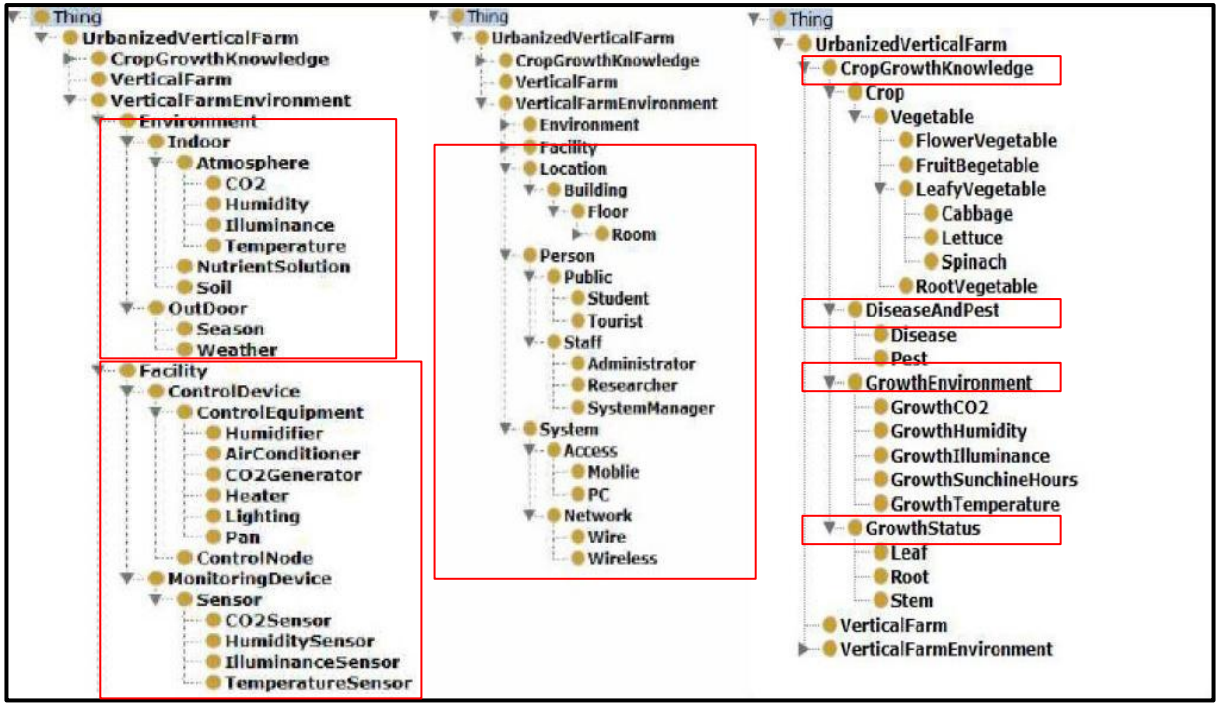
Hydrological Ontology for Flood stage detection

- Hydrological sensor web ontology
- Precipitation from weather Stations
- Water level from hydrological stations
- Stage of flood - mitigation, preparedness, response, recovery
- W3C SSN ontology extension



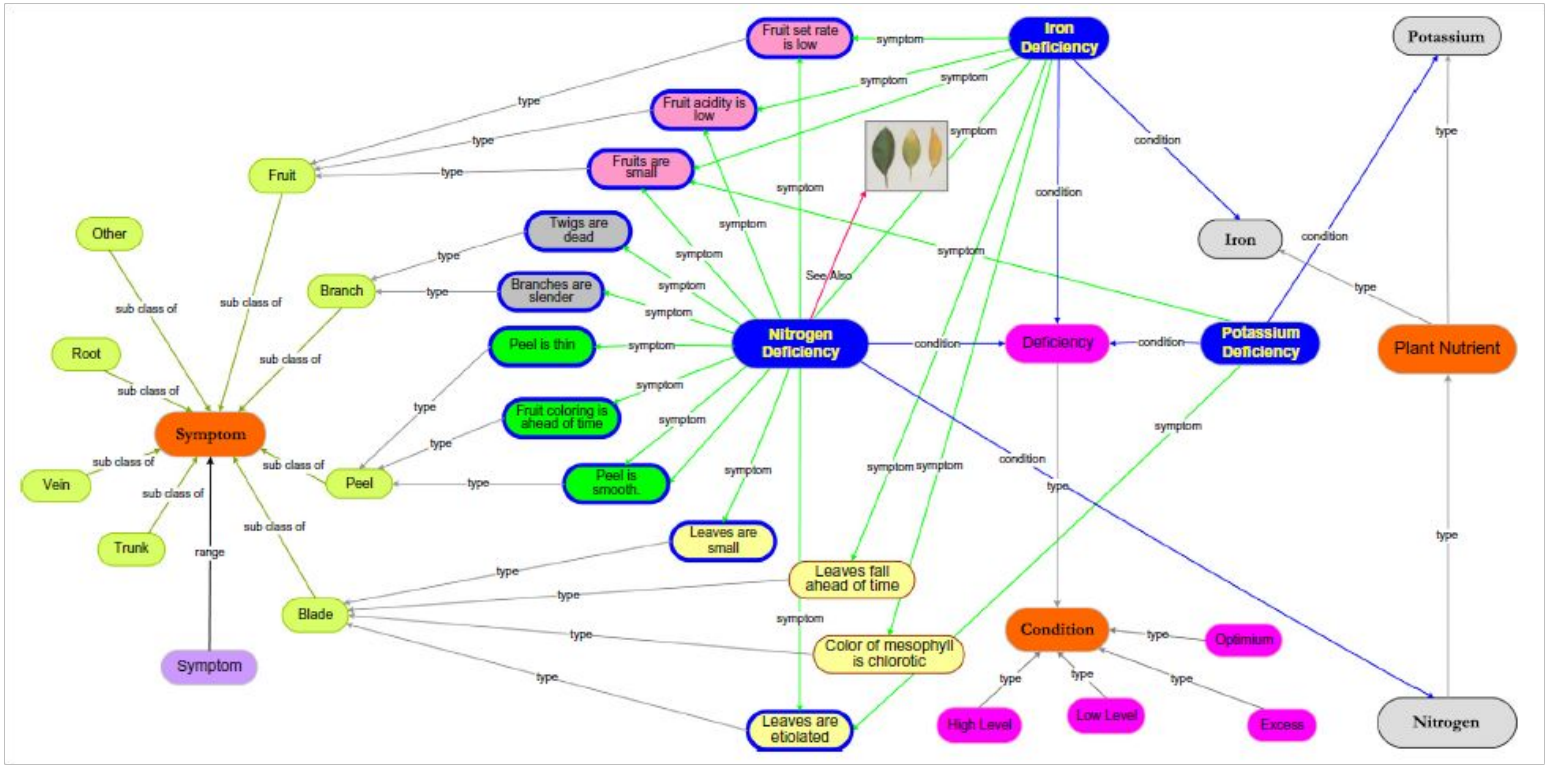
Ref: Wang et al., A hydrological sensor web ontology based on the ssn ontology: A case study for a flood (2017)

Agricultural Ontology Model



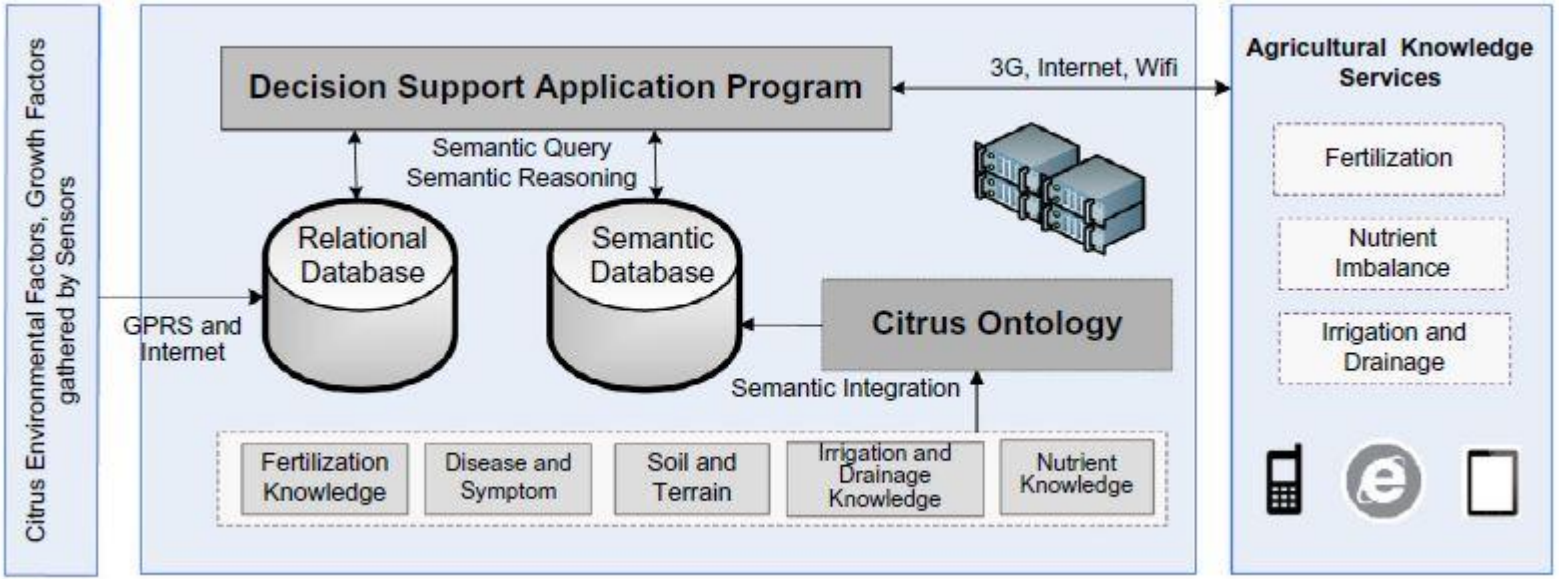
Ref: Kim et al.,A study of an agricultural ontology model for an intelligent service in a vertical farm (2013)

Knowledge base for Hilly citrus production



Ref: Wang et al., An ontology-based approach to integration of hilly citrus production knowledge (2015)

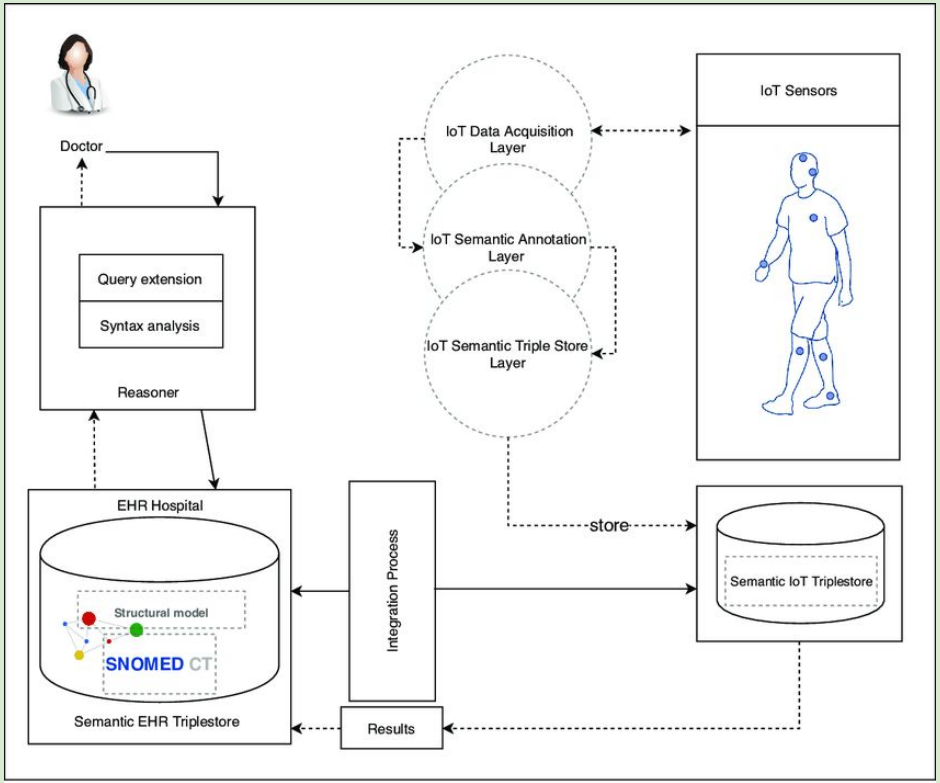
Services using knowledge base



Ref: Wang et al., An ontology-based approach to integration of hilly citrus production knowledge (2015)

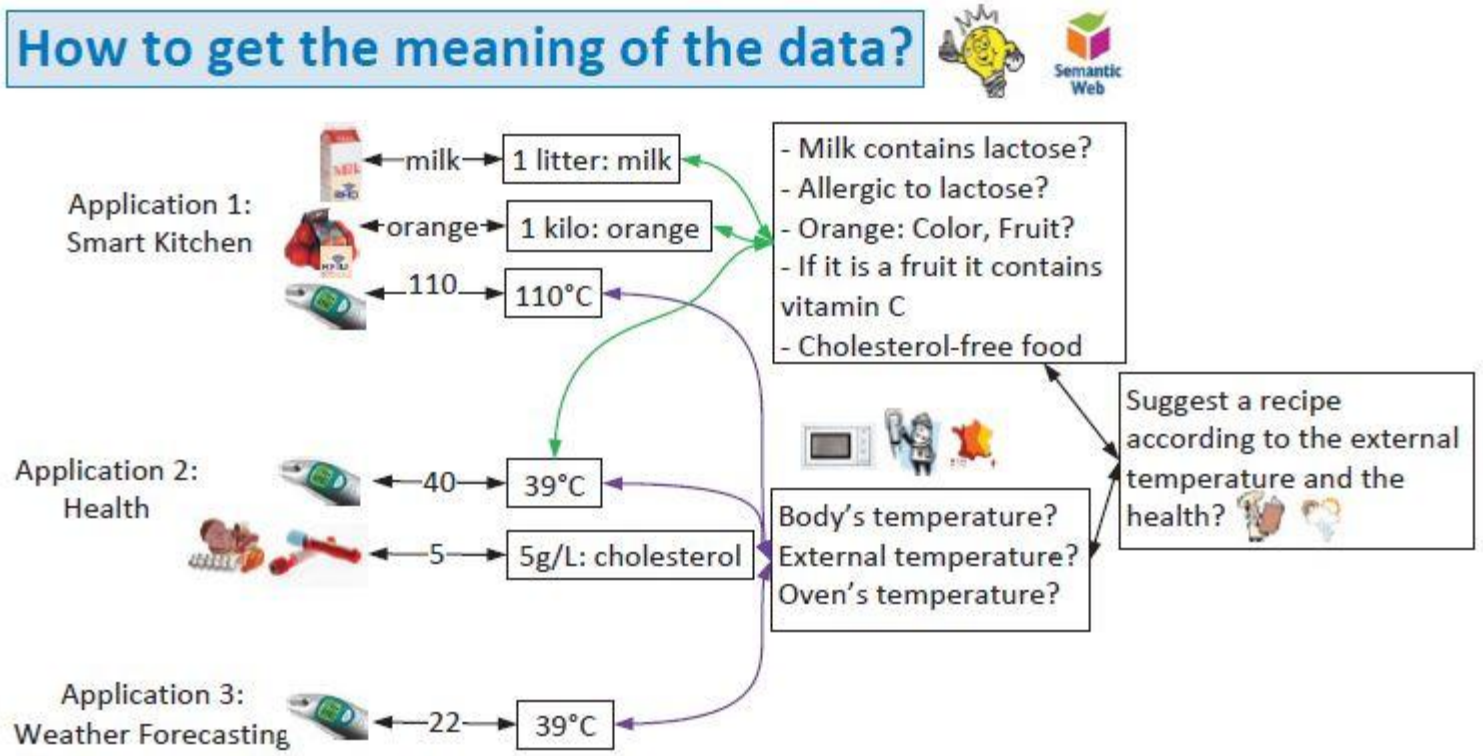
Middleware for integrating Healthcare IoT systems

- Integrate Electronic Health records and IoT sensor data
- SNOWMED-CT Systematized Nomenclature of Medicine-Clinical Terms
- W3C SSN ontology
- OWL Time, Geo Ontology



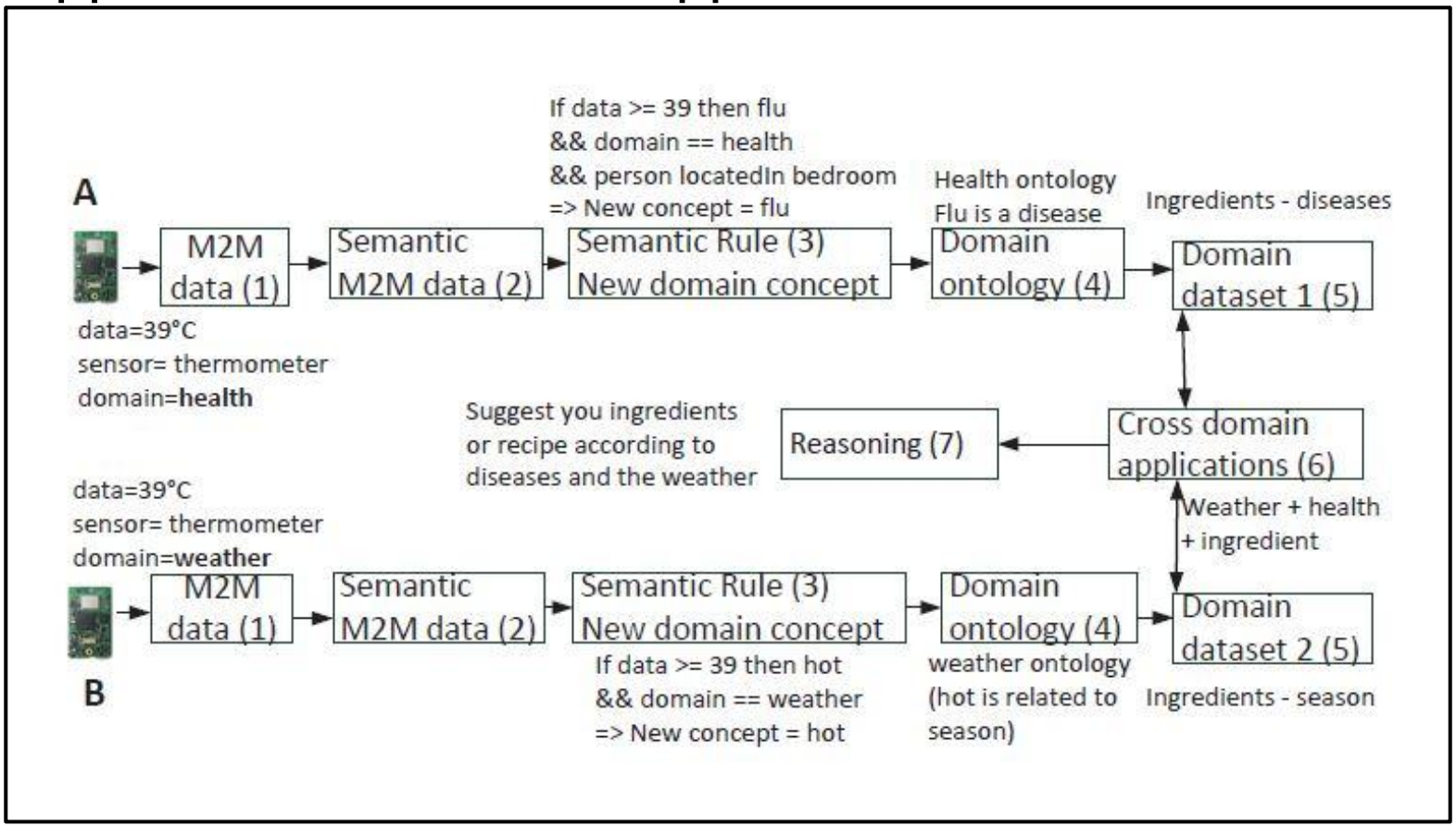
Ref: Abdullah Alamri, Ontology middleware for integration of iot healthcare information systems in ehr systems (2018)

Cross domain Application Scenario



Ref: A. Gyrard et al., Enrich machine-to-machine data with semantic web technologies for cross-domain applications (2014)

M3 Approach for Semantic Applications



Ref: A. Gyrard et al., Enrich machine-to-machine data with semantic web technologies for cross-domain applications (2014)

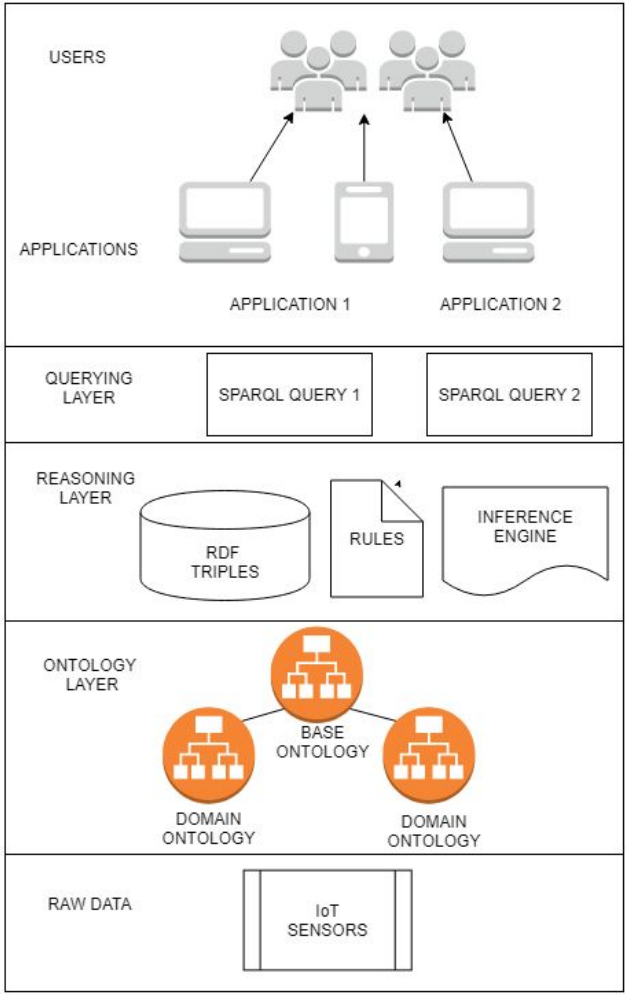
Research Objectives

- Develop an interoperable and scalable semantics framework to build cross domain IoT applications
- Design a basic ontology using standardised W3C SSN to be used for any IoT application
- Develop application specific ontologies and rules
- Semantics-driven IoT application for agriculture

Methodology and Implementation

Semantics-Driven Application Architecture

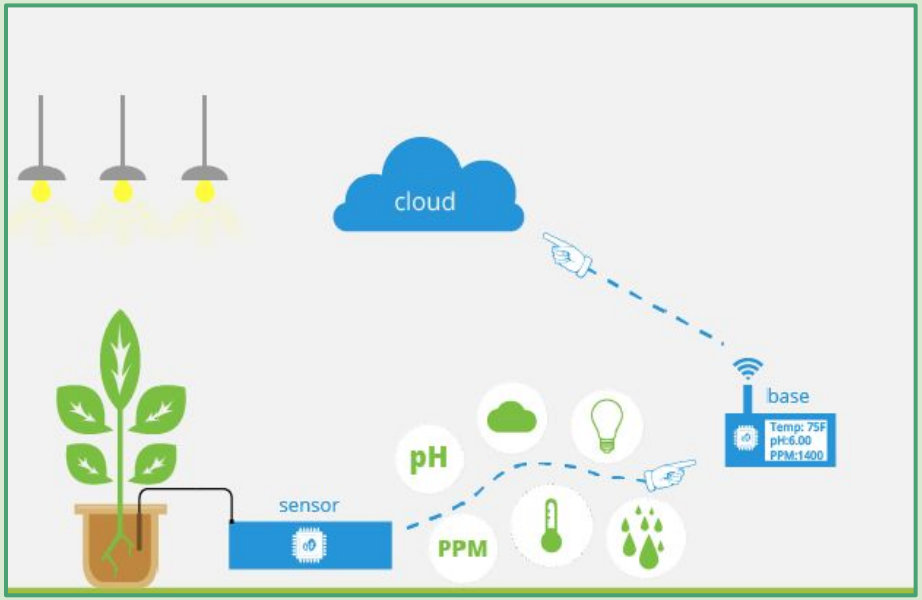
Semantic MiddleWare



IoT application for agriculture

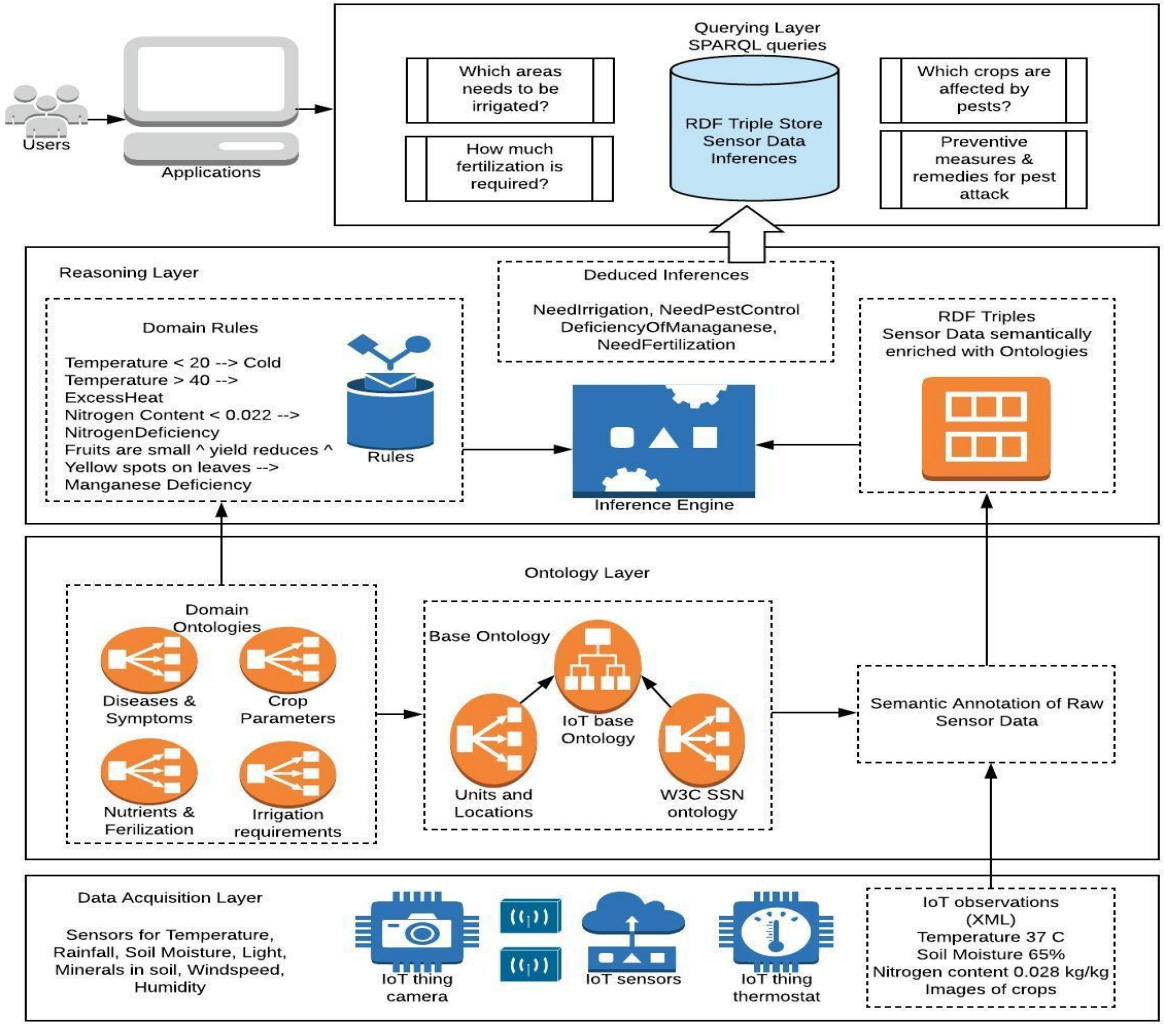
- Crop yield Analysis
- Water Stress
- Fertilization
- Field Monitoring
- Diagnosis of Diseases

★ SenseQube Agro-Meteorological data
★ Citrus Orchards
★ Plant Fungal Diseases caused by Phytophthora



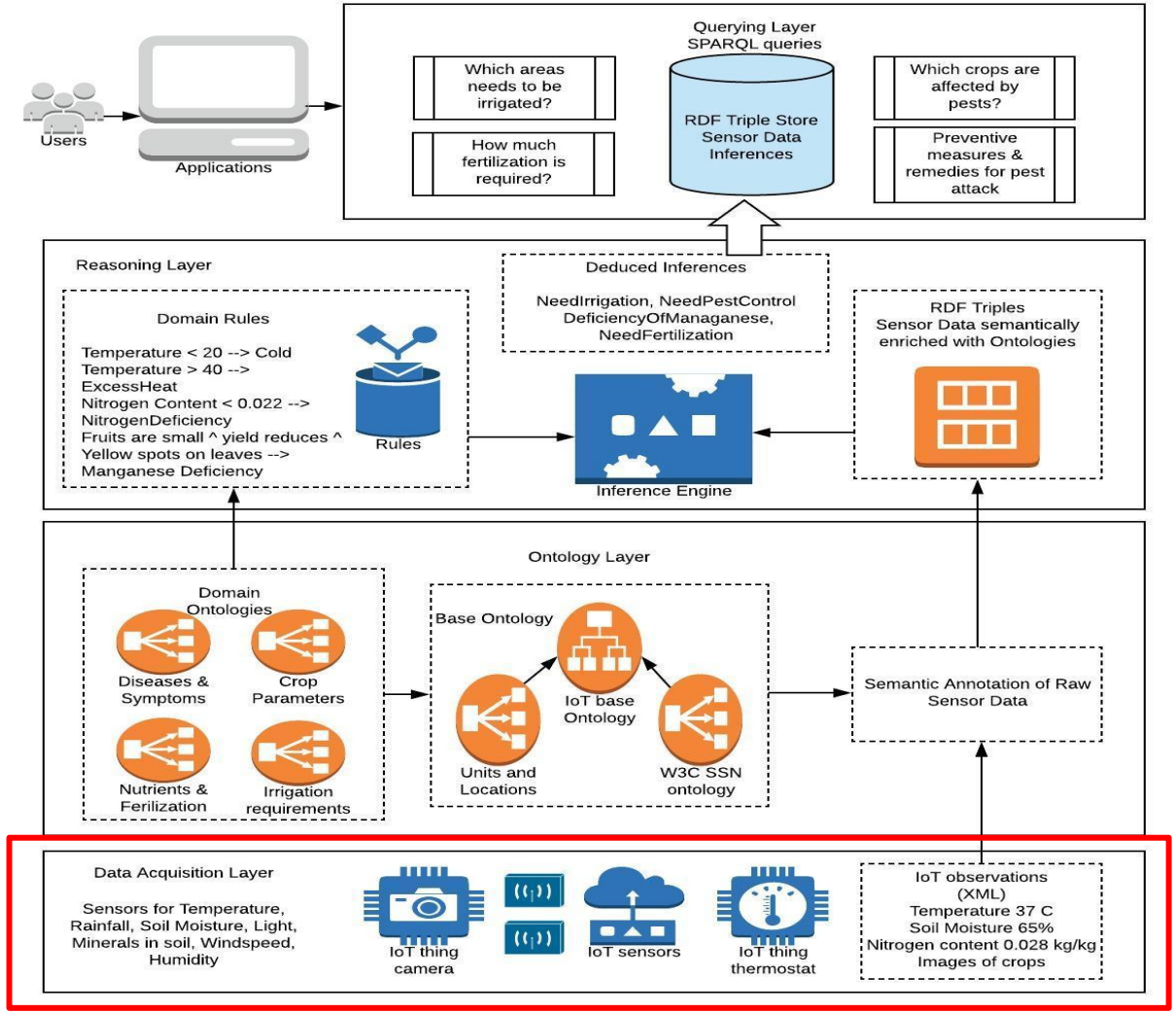
Source: https://qph.fs.quoracdn.net/main-qimg-cd4c2b8d4a1e8d6fdfe5e3d36e684355.webp

Overall Architecture



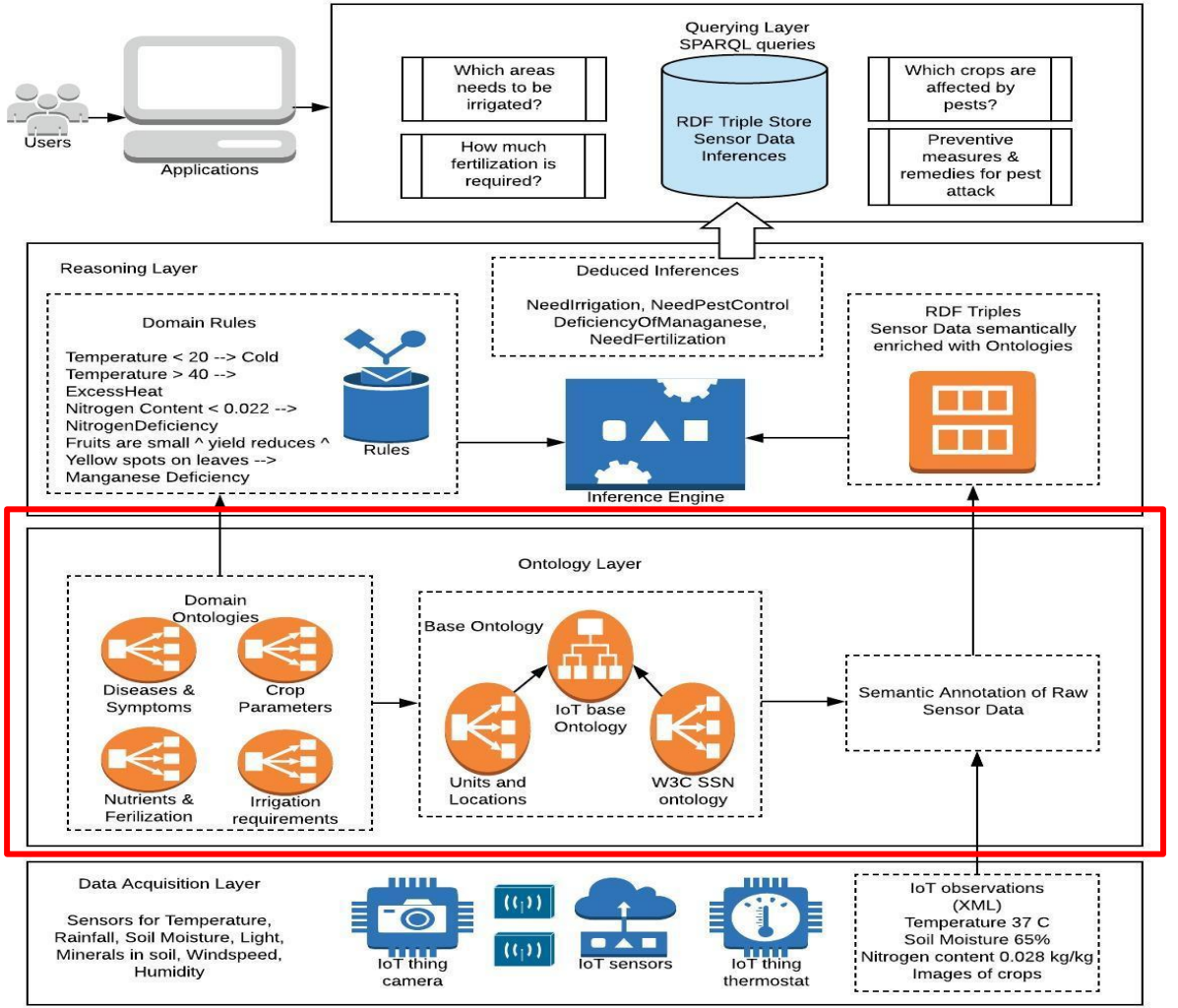
- Sensors deployed on field - SenseQube
- Environmental parameters
- Crop parameters

Data Acquisition



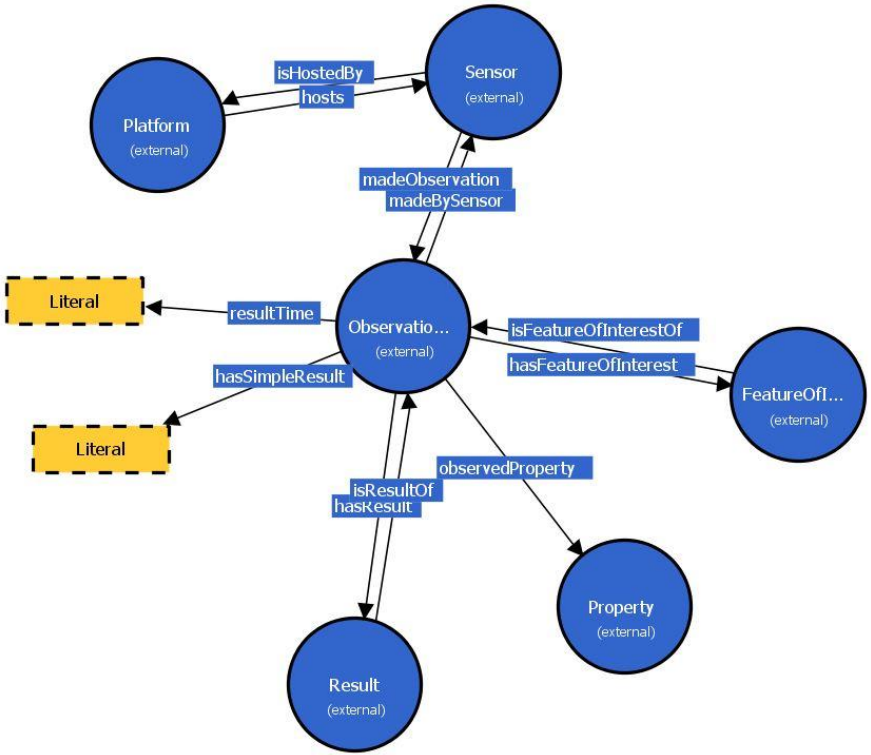
- Base Ontology
- Domain specific Ontologies
- Semantically annotate data

Ontology Layer



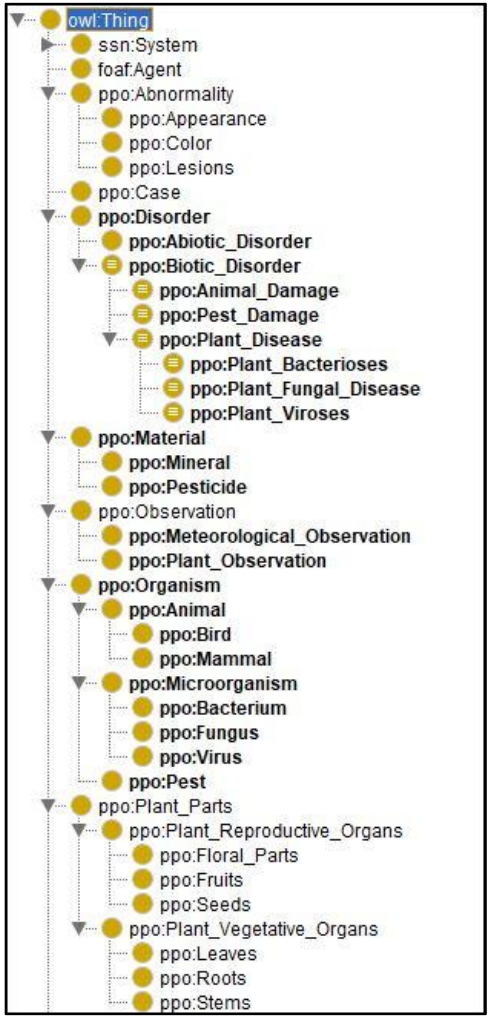
SSN

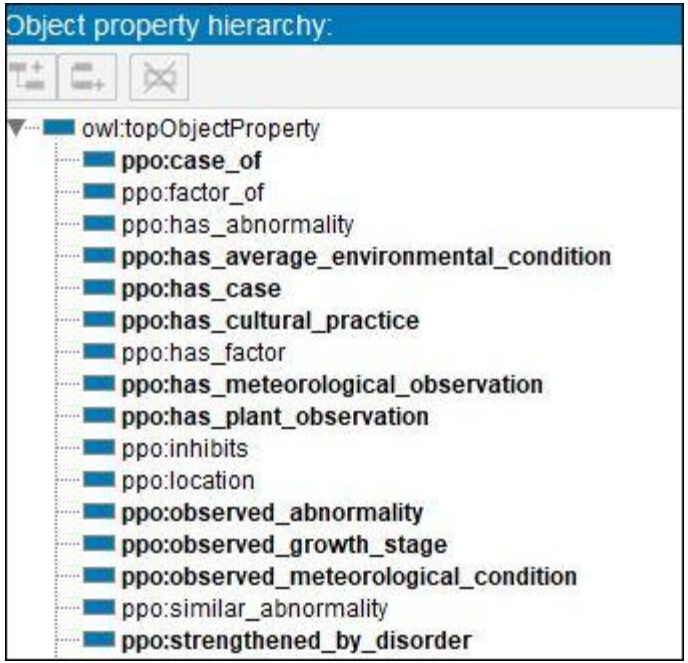
- W3C recommended
- SOSA - Sensor, Observation, Sampler and Actuator
- SSN - Semantic Sensor Network
- SenseQube Modelling
  - SenseQube
  - Sensors
  - Locations
  - Observations



PPO ontology

- Plant Protection Ontology
- Disorders
- Plant Fungal Diseases
- Phytophthora spp.
- Plant parts that the diseases will affect
- Symptoms or Abnormalities

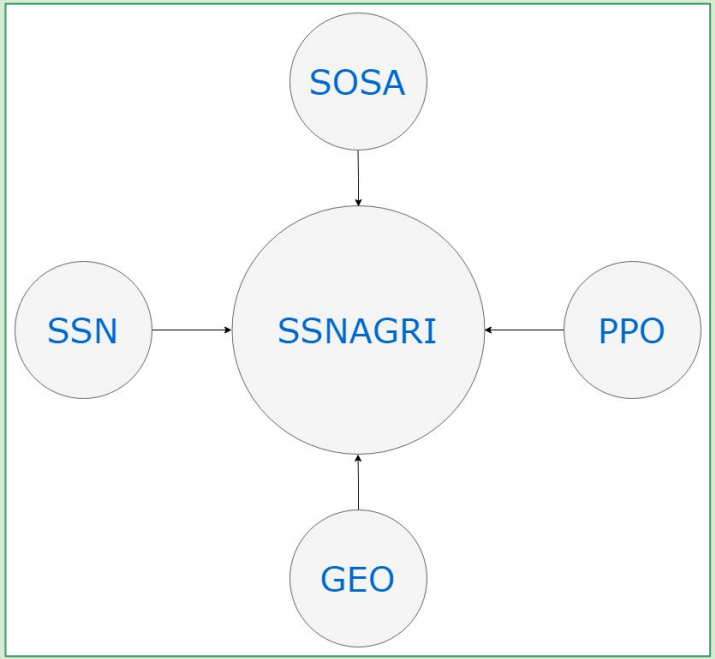




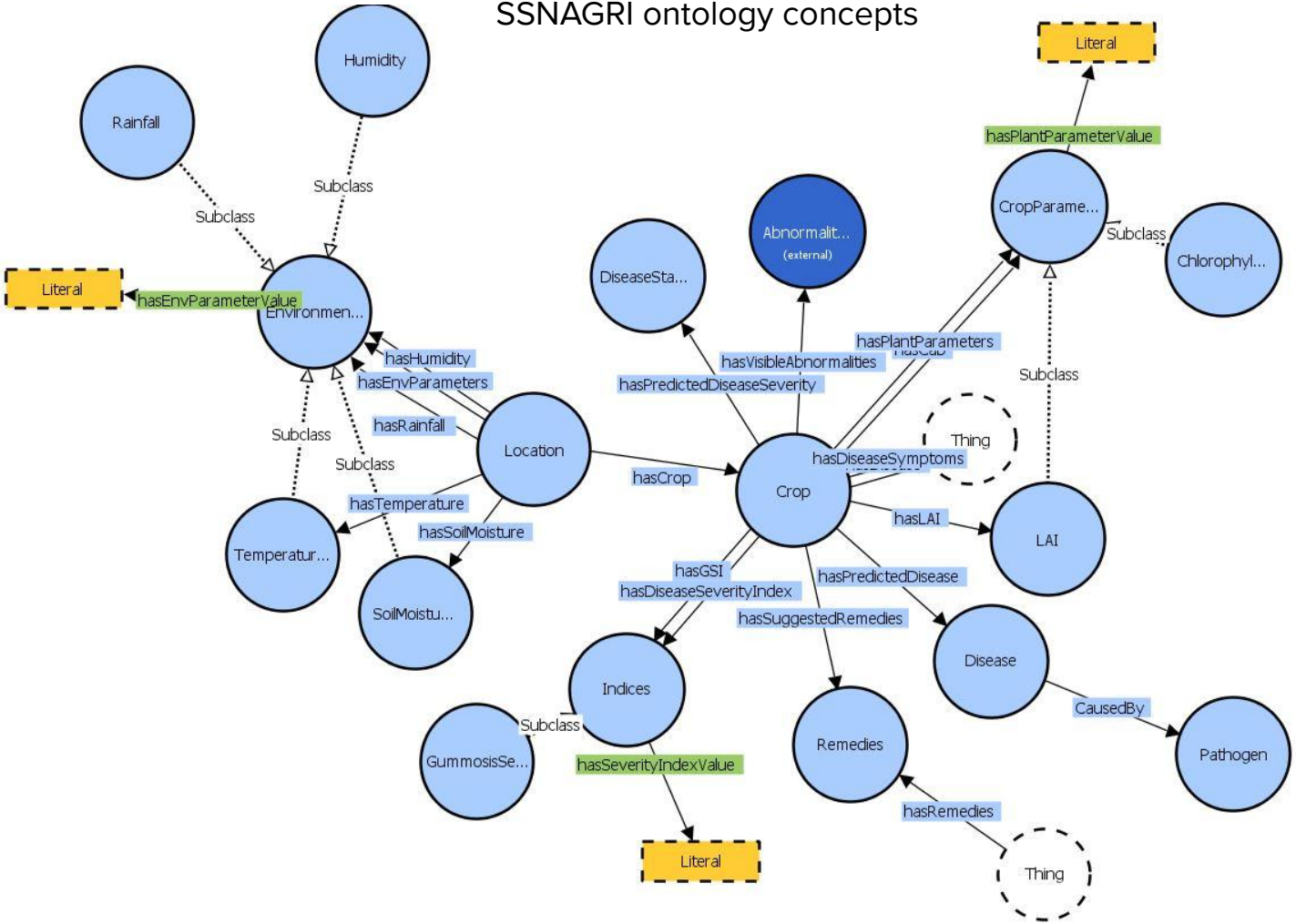
Ref: R Gade and R Lad. Biological management of major citrus diseases in central india-a review.

SSNAGRI

- SSN for Agriculture Ontology
- Sensors and Observations
- Crop, Location, Environmental parameters
- Plants, Diseases, Symptoms, Causes
- Individuals for modelling the scenario

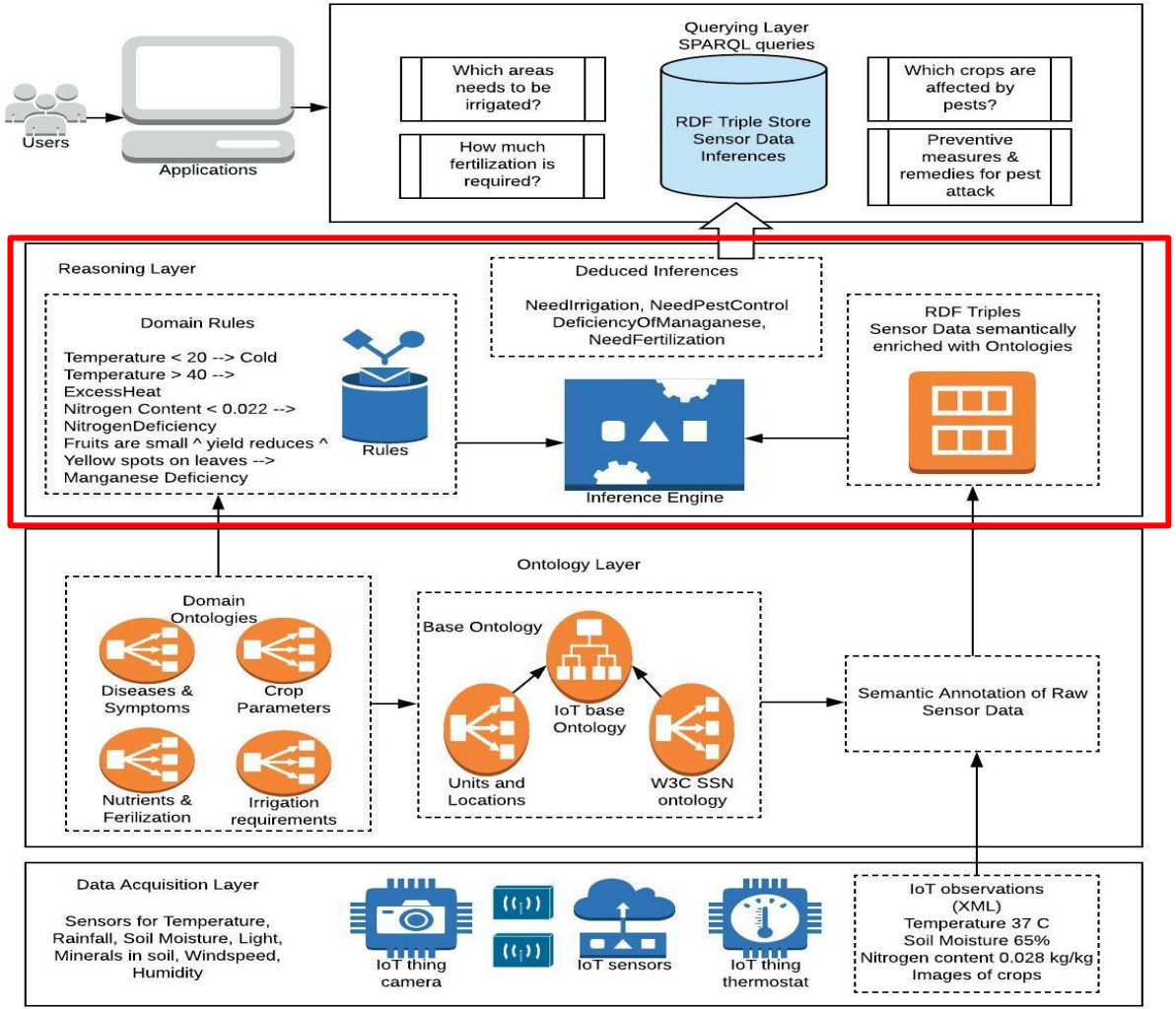


SSNAGRI ontology concepts



Reasoning Layer

- Rules for deriving real world concepts
- RDF Triples
- Inference Engine



Rules

- Heat levels from temperature (Extreme heat, heat, normal, cold)
- Water content from soil moisture (Dry, Medium, Wet)
- Disease severity (Severe, Mild, Onset, Normal)
- Minerals content (Excess, High, Optimum, Low, Deficiency)
- Pests (Present, Absent)
- Irrigation (Required, Not Required, Status)

Semantic Web Rule Language (SWRL) is used for formulating the rules.

SWRL Rule for Gummosis Severity Index

$$GS = -0.0666*T -0.0033*R -0.0148*LAI + 0.2134*Cab + 3.2631$$

Name

GSI Rule

Comment

Gummosis Severity from Parameters

Status

Ok

```
ssnagri:Crop(?cr) ^ ssnagri:isLocatedIn(?cr, ?loc) ^ ssnagri:hasTemperature(?loc, ?t) ^ ssnagri:hasRainfall(?loc, ?r) ^
ssnagri:hasCab(?cr, ?c) ^ ssnagri:hasLAI(?cr, ?l) ^ ssnagri:hasGSI(?cr, ?g) ^ ssnagri:Temperature(?t) ^
ssnagri:Rainfall(?r) ^ ssnagri:Chlorophyll(?c) ^ ssnagri:LAI(?l) ^ ssnagri:hasEnvParameterValue(?t, ?t1) ^
swrlb:multiply(?t2, ?t1, -0.0666) ^ ssnagri:hasEnvParameterValue(?r, ?r1) ^ swrlb:multiply(?r2, -0.0033, ?r1) ^
ssnagri:hasPlantParameterValue(?l, ?l1) ^ swrlb:multiply(?l2, -0.0148, ?l1) ^ ssnagri:hasPlantParameterValue(?c, ?c1) ^
swrlb:multiply(?c2, 0.2134, ?c1) ^ swrlb:add(?gs, ?t2, ?r2, ?l2, ?c2, 3.2631) -> ssnagri:hasSeverityIndexValue(?g, ?gs)
```

Ref: Badnakhe et al., Evaluation of Citrus Gummosis disease dynamics and predictions with weather and inversion based leaf optical model (2018)

SWRL Rule for Disease Prediction using GSI

Name

Disease Offset disease rule

Comment

Disease Severity from GSI

Status

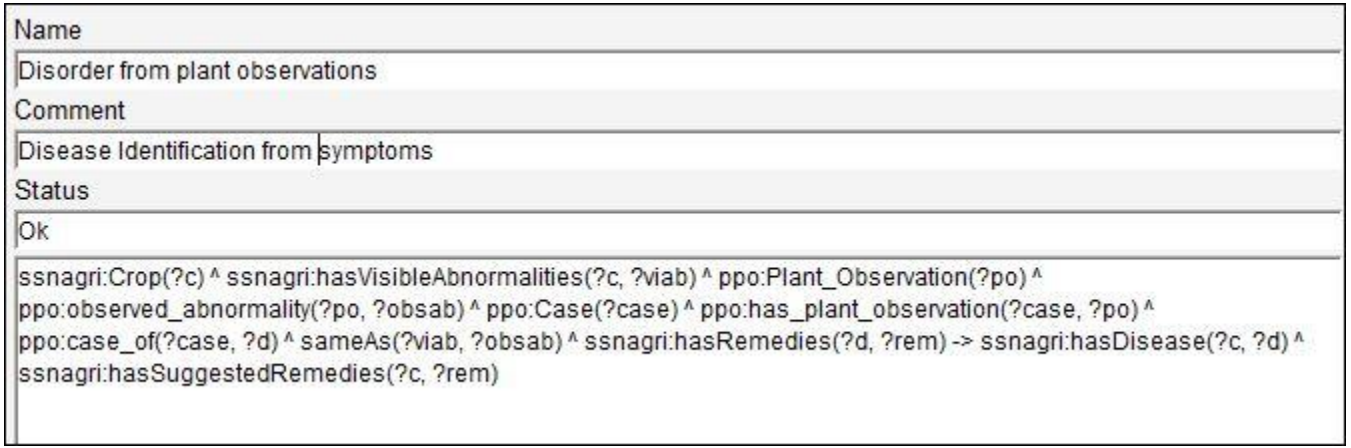
Ok

```
ssnagri:Crop(?c) ^ ssnagri:hasDiseaseSeverityIndex(?c, ?i) ^ ssnagri:hasSeverityIndexValue(?i, ?v) ^
swrlb:greaterThan(?v, 1) ^ swrlb:lessThan(?v, 2) -> ssnagri:hasPredictedDiseaseSeverity(?c, ssnagri:OffsetOfDisease) ^
ssnagri:hasPredictedDisease(?c, ssnagri:Gummosis)
```

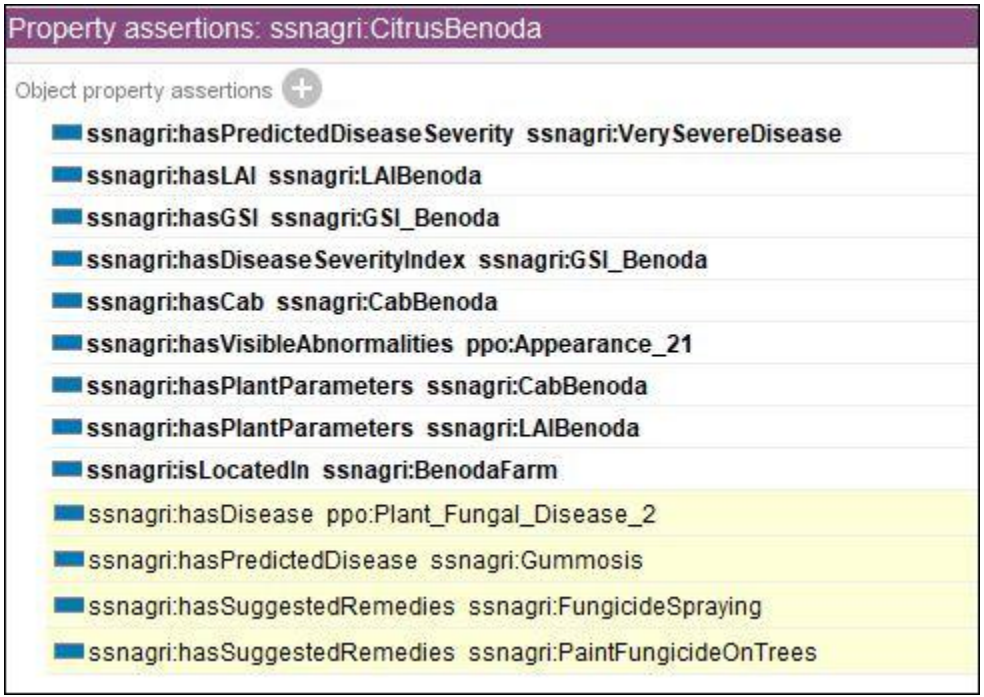
Ref: Badnakhe et al., Evaluation of Citrus Gummosis disease dynamics and predictions with weather and inversion based leaf optical model (2018)

SWRL Rule for Disease Identification from Plant Observations

- Plant Disease and the abnormalities caused are defined in the PPO ontology
- If SSNAGRI Crop Instance has similar Visible Abnormalities, then derive that the Crop has the disease

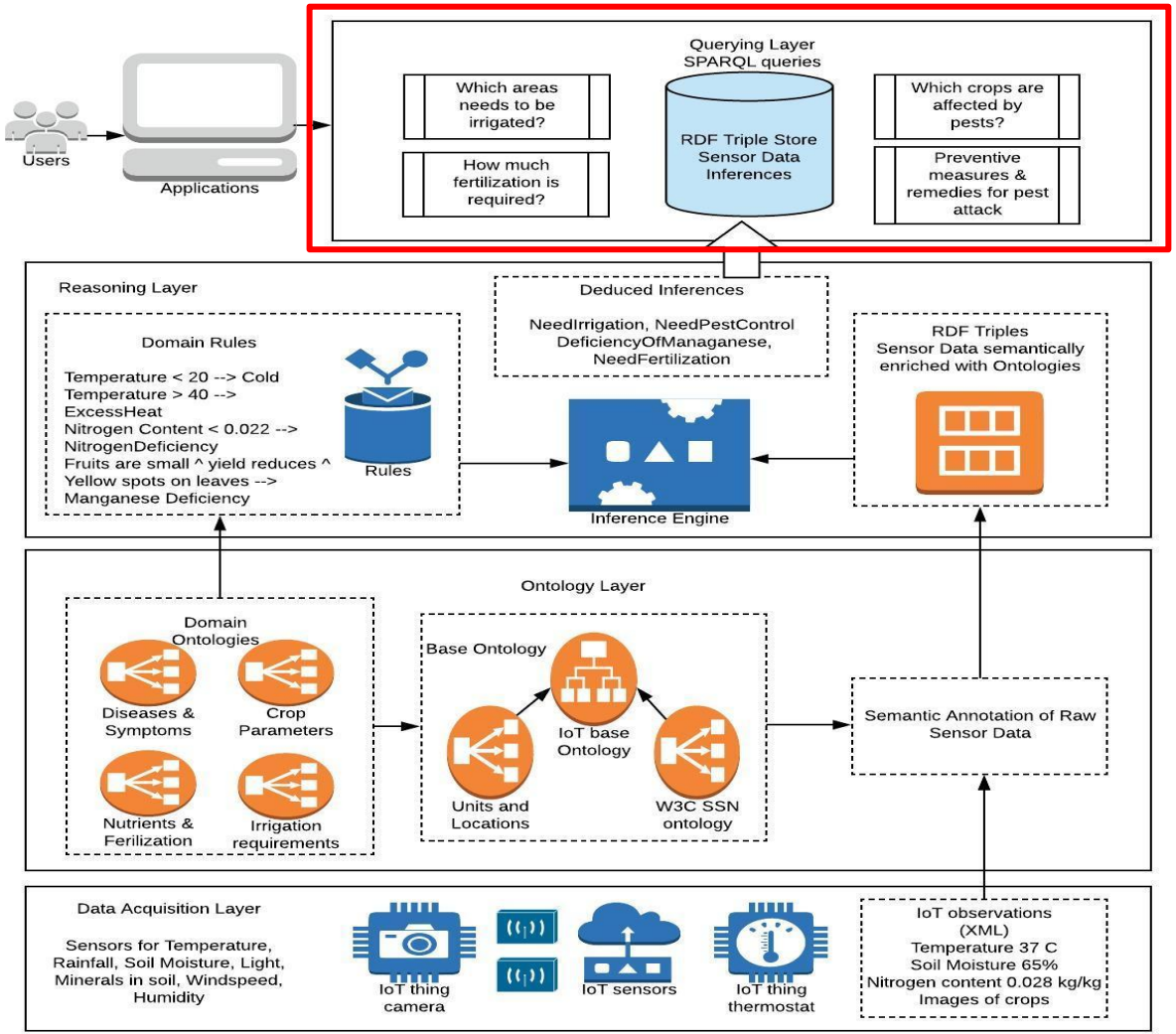


Inferences


Property assertions: ssnagri:CitrusBenoda
Object property assertions
ssnagri:hasPredictedDiseaseSeverity ssnagri:VerySevereDisease
ssnagri:hasLAI ssnagri:LAIBenoda
ssnagri:hasGSI ssnagri:GSI_Benoda
ssnagri:hasDiseaseSeverityIndex ssnagri:GSI_Benoda
ssnagri:hasCab ssnagri:CabBenoda
ssnagri:hasVisibleAbnormalities ppo:Appearance_21
ssnagri:hasPlantParameters ssnagri:CabBenoda
ssnagri:hasPlantParameters ssnagri:LAIBenoda
ssnagri:isLocatedIn ssnagri:BenodaFarm
ssnagri:hasDisease ppo:Plant_Fungal_Disease_2
ssnagri:hasPredictedDisease ssnagri:Gummosis
ssnagri:hasSuggestedRemedies ssnagri:FungicideSpraying
ssnagri:hasSuggestedRemedies ssnagri:PaintFungicideOnTrees

Query Layer

SPARQL queries on knowledge base to retrieve observed values, detected events, inferences



Crops with Diseases

Snap SPARQL Query:

```
PREFIX ppo: <http://www.ontologies.com/ppo#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX sosa: <http://www.w3.org/ns/sosa/>
PREFIX ssn: <http://www.w3.org/ns/ssn/>
PREFIX ssnagri: <http://www.semanticweb.org/csreiitb/ontologies/ssnagri#>
SELECT ?Crop ?Location  ?Disease
            WHERE {
            ?Crop a ssnagri:Crop .
            ?Crop ssnagri:hasDisease ?d .
            ?d ppo:name ?Disease .
            ?Crop ssnagri:isLocatedIn ?Location .
}
```

Execute

?Crop	?Location	
ssnagri:CitrusCrops_2	ssnagri:NagziriFarm	Brown fruit rot^^xsd:string
ssnagri:CitrusCrops_2	ssnagri:NagziriFarm	Root rot^^xsd:string
ssnagri:CitrusBenoda	ssnagri:BenodaFarm	Foot rot^^xsd:string

Diseases which affects stem

Snap SPARQL Query:

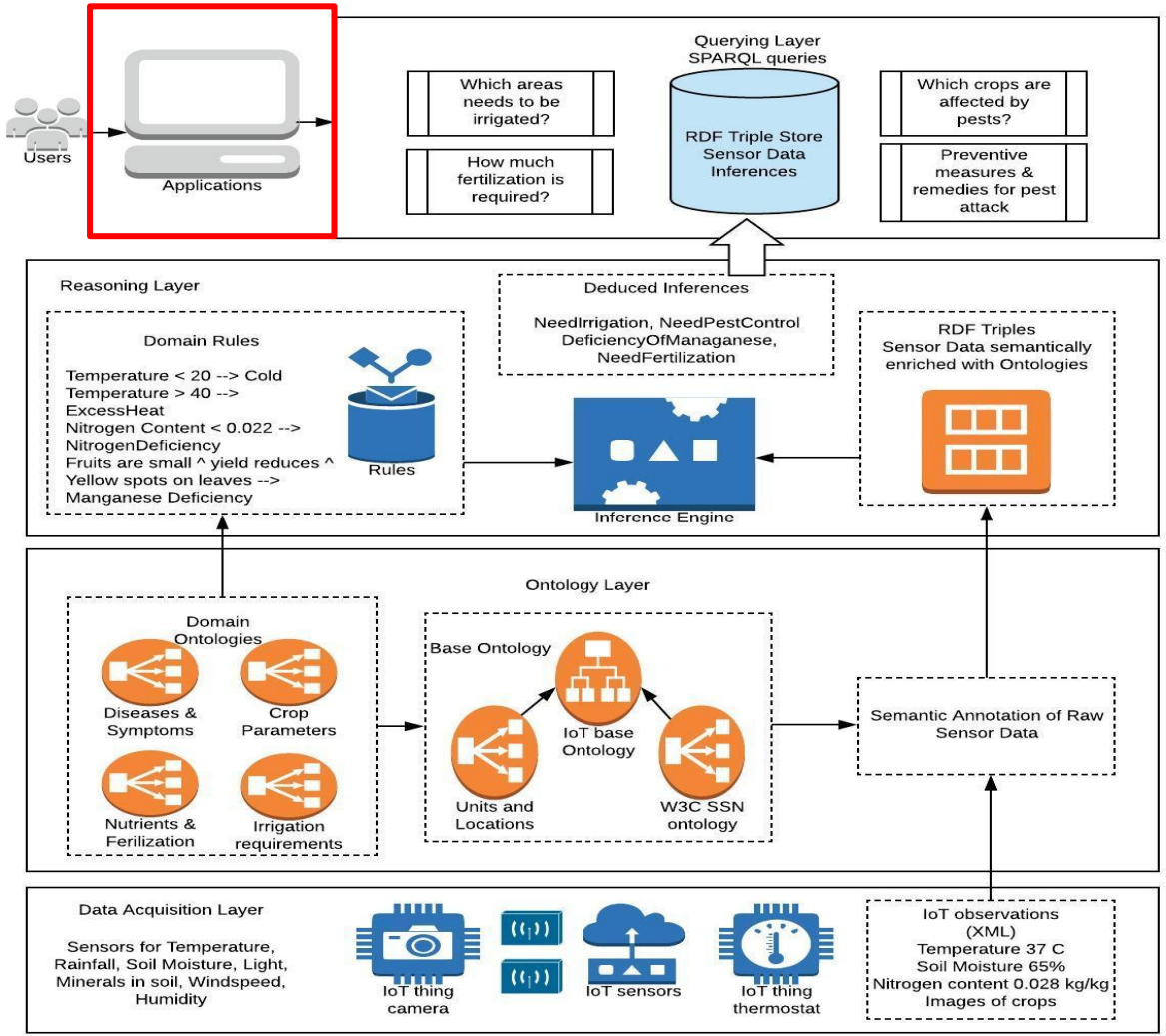
```
PREFIX ppo: <http://www.ontologies.com/ppo#>
SELECT ?causedBy ?disease ?symptom  ?part
WHERE {
            ?d a ppo:Disorder .
            ?d ppo:name ?disease .
            ?d ppo:has_case ?c .
            ?c ppo:has_plant_observation ?p .
            ?p ppo:observed_abnormality ?ab .
            ?ab ppo:value_category ?symptom .
            ?d ppo:has_factor ?f .
            ?f ppo:scientific_name ?causedBy .
            ?ab ppo:location ?loc .
            ?loc ppo:name ?part .
            FILTER CONTAINS(LCASE(?part),"stem").
             }
```

Execute

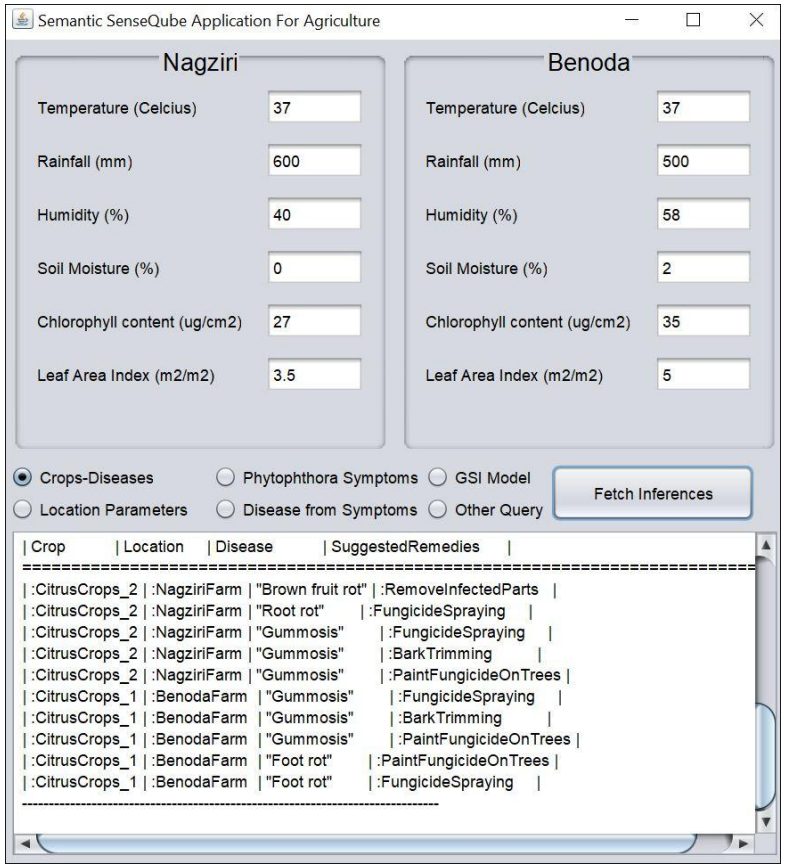
?causedBy	?disease	?symptom	
Phytophthora spp.^^xsd:string	Gummosis^^xsd:string	Lesions around circumference of trunk^^xsd:string	Stems^^xsd:string
Phytophthora spp.^^xsd:string	Foot rot^^xsd:string	Infection of young shoot near the ground^^xsd:string	Stems^^xsd:string

Applications

- User Interface
- Irrigation map
- Fertilization requirements
- Growth status of plants
- Diseases and suggested remedies



Semantic SenseQube Application

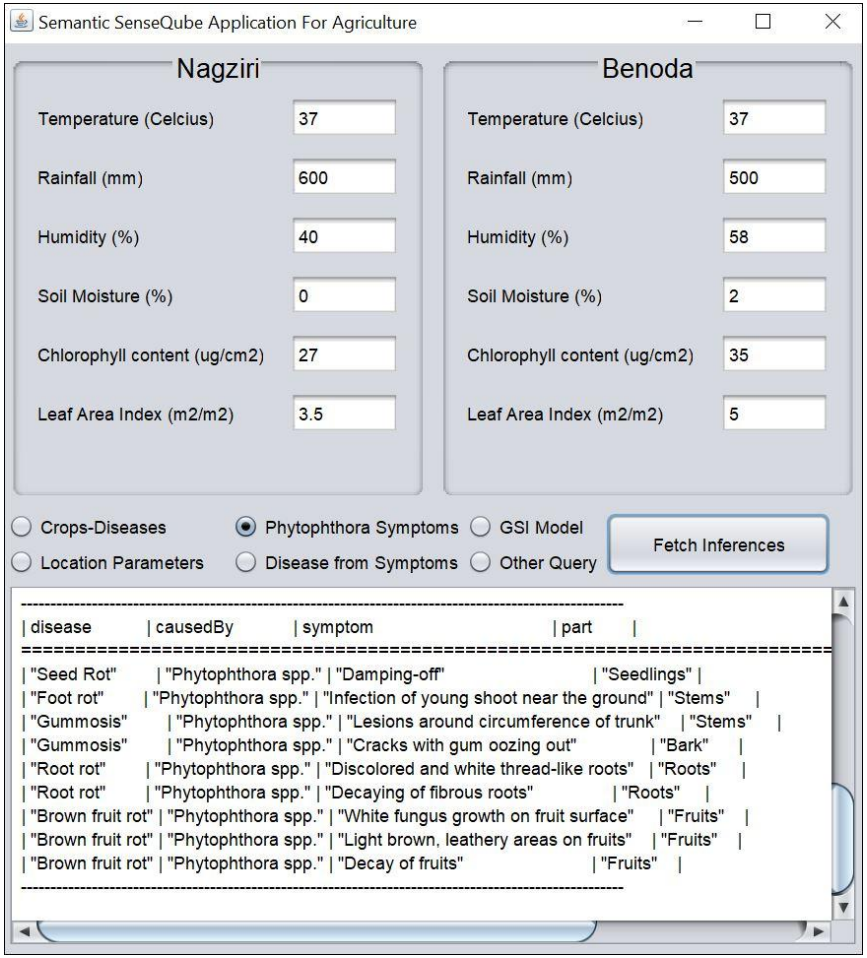


SenseQube deployed at Citrus Orchards
Provide Sensor Values as Input

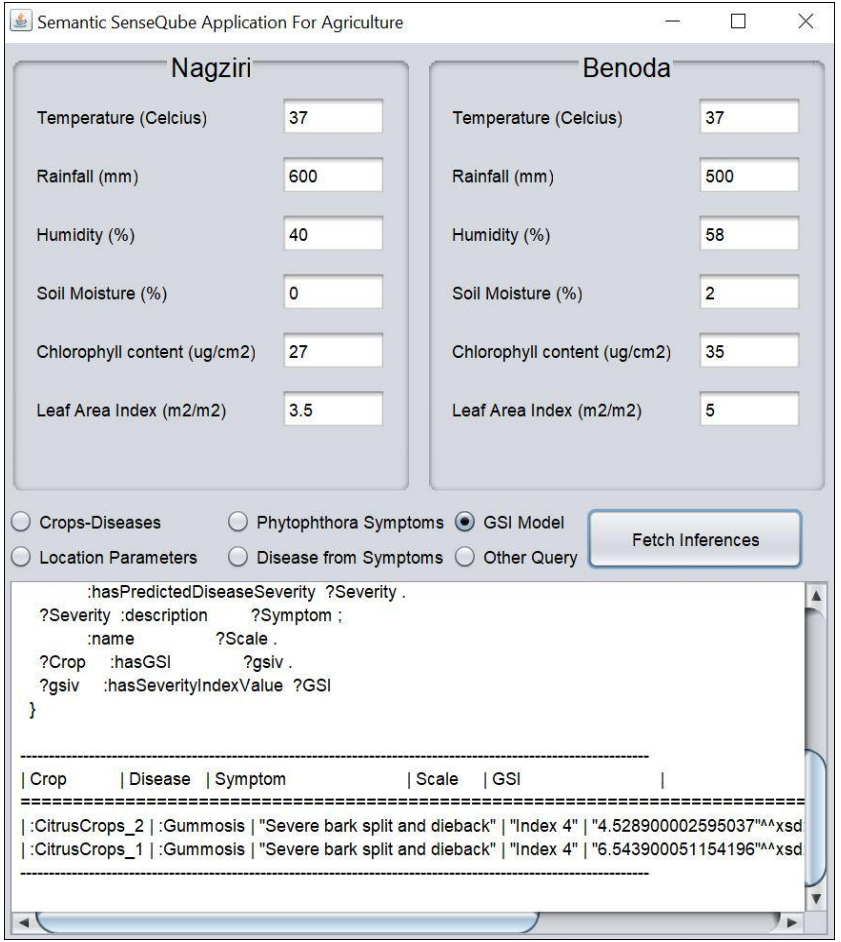
Choose a query
Then, Fetch Inferences

Results will be displayed in this panel

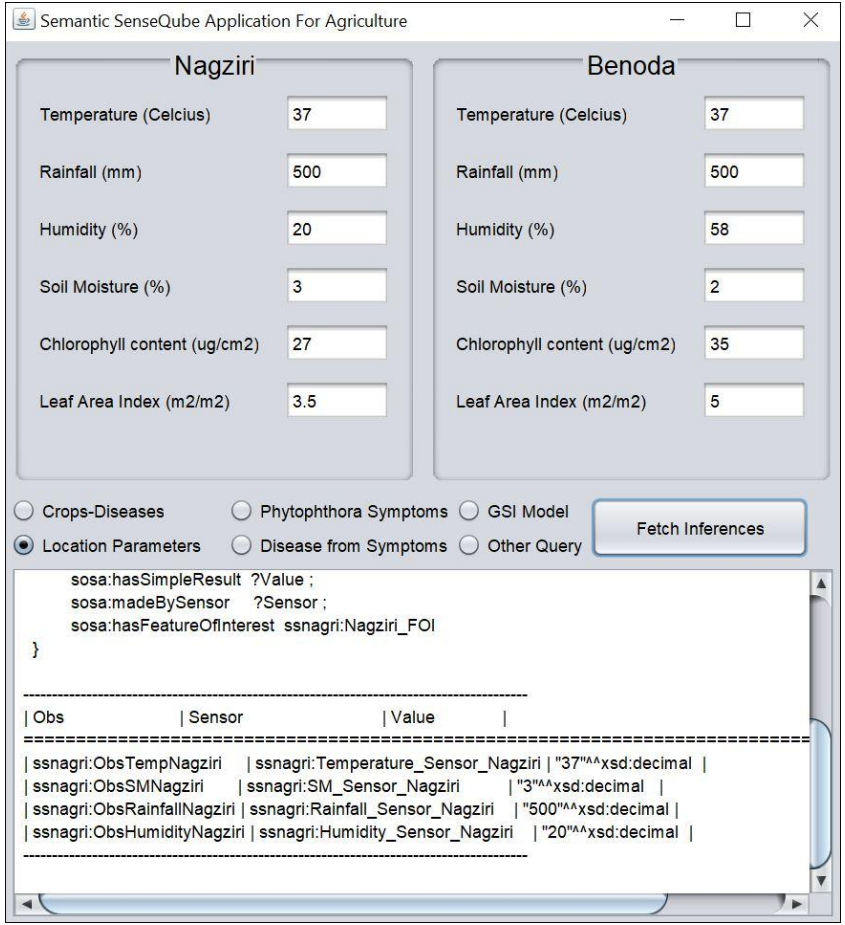
Symptoms of Phytophthora fungal diseases



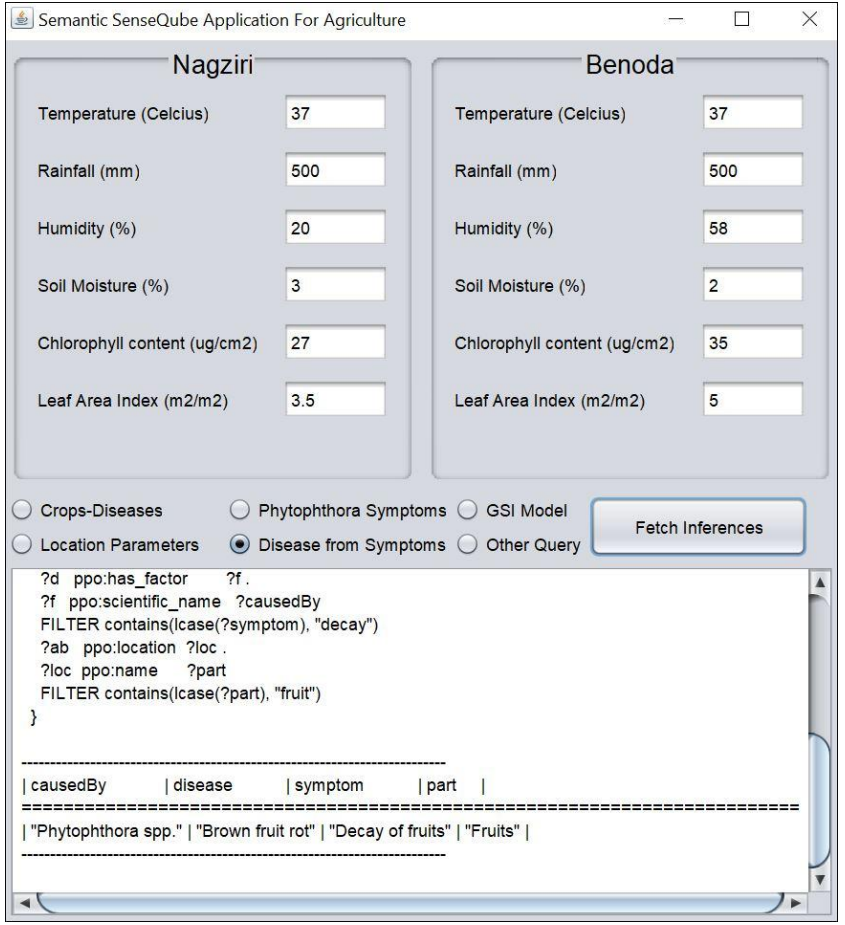
Gummosis Severity Index Value



Weather Parameters of a Location



Diseases which cause decay in fruits

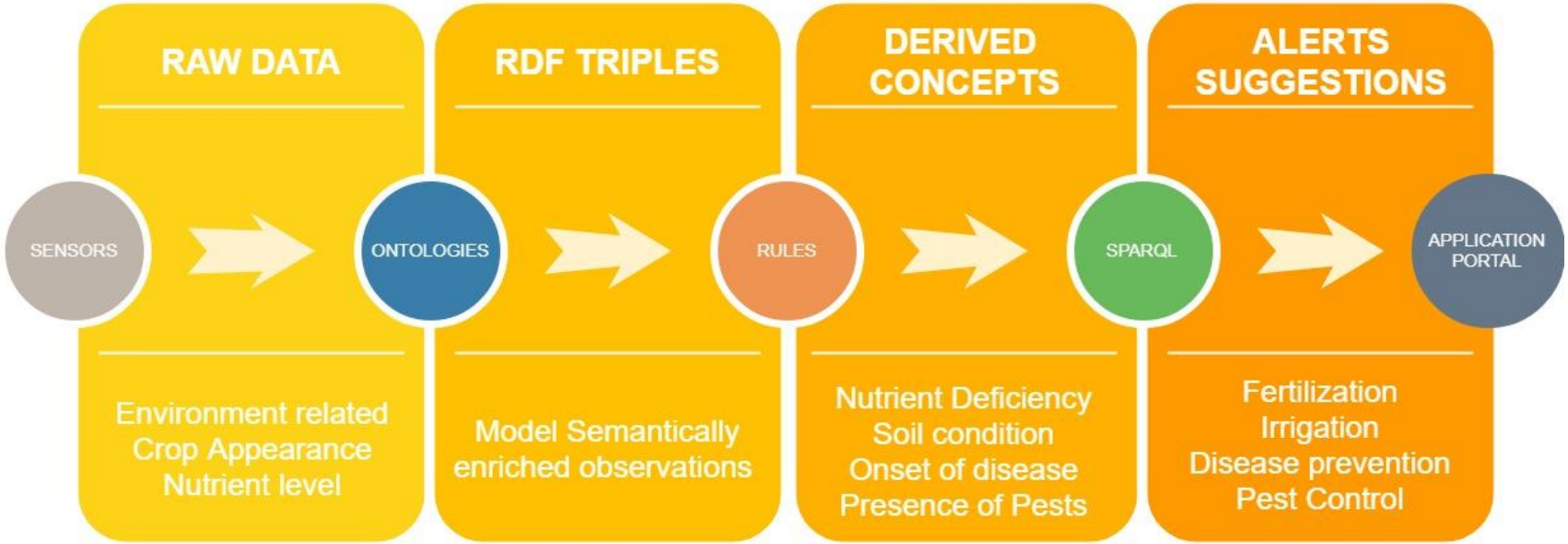


Software and Tools

- Ontology creation and validation - Protege, VOWL plugin
- Rules formulation - SWRL plugin in Protege
- User interface for application - Swing
- Semantic annotation of data - JAVA
- Modelling the RDF triples, ontologies - Apache Jena
- Reasoning - Pellet, Generic Rule Reasoner
- SPARQL for querying

Summary

- ★ Ontology design with W3C SSN, Crop, Disease concepts
- ★ Knowledge based reasoning
- ★ Semantic Application Framework



Future Work

- ★ Other Applications - Irrigation, Fertilization etc.
- ★ Extend the ontologies with detailed domain knowledge
- ★ Integration with SenseQube for Real time processing
- ★ Triple data store

Thank You